MIL-O-60863A (AR)
7 March 1986
SUPERSEDING
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15 December 1967

MILITARY SPECIFICATION

OXAMIDE

This Specification is approved for use within the US Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of oxamide for use in explosives as follows:

Type I - Oxamide meeting the particle size requirements given in 3.1.
Type II - Oxamide meeting the granulation requirements given in 3.1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves, Standard for Testing Purposes
PPP-C-301 - Chemicals, Dry and Paste, Packaging and Packing of
PPP-D-723 - Drums, Fiber

MILITARY

MIL-D-6054 - Drums, Metal - Shipping and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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STANDARDS

MIL-STD-105	-	Sampling Procedures and Table for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-650	-	Explosive, Sampling, Inspection and Testing
MIL-STD-1218	-	ACS Chemicals
MIL-STD-1235	-	Single and Multi Level Continuous Sampling Procedures and Tables for Inspection by Attributes

(Copies of Specifications, standards, drawings and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following other publications form a part of this specification to the extent specified herein. The issue of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation E300	-	Recommended Practice for Sampling Industrial Chemicals
ASTM Designation D1193	-	Reagent Water
ASTM Designation D2905	-	Statement of Number of Specimens Required to Determine the Average Quality of a Textile Material

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical requirements. Oxamide shall conform to the applicable chemical and physical requirements specified in Table I when tested as specified in the applicable paragraph.

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TABLE I. Chemical and physical requirements for oxamide

TEST	TYPE I	TYPE II	TEST PARAGRAPH
Purity, percent, min	99.0	99.0	4.6.1
Moisture content, percent, max	0.1	0.1	4.6.2
Acidity, as oxalic acid, percent, max	0.05	0.05	4.6.3 4.6.4
Residue, percent, max	0.05	0.05	4.6.5
Particle size, microns	4 ± 2		4.6.6
Granulation, percent			4.6.7
Thru US Standard Sieve No. 30		100.0	
Thru US Standard Sieve No. 100, min		80.0	

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.1.1).

3.3 Workmanship. The manufacturer shall use procedures and controls which assure the oxamide produced does not contain foreign materials such as dirt, rust, paint or metal chips, etc. Compliance with this requirement shall be as specified in 4.6.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any suitable facilities, acceptable to the government for the performance of the inspections. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein.

4.2 Classification of inspections. The following inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Packaging inspection (see 4.4.2 and 5).

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of three kilograms (6.6 pounds) of the composition obtained by sampling as described in 4.5.1. The samples shall be obtained from a production batch which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to be performed. The sample will be subjected by the government to any or all of the examinations and tests specified in Table II of this specification and any or all requirements of the applicable drawings.

4.3.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements.

TABLE II. First article inspection

EXAMINATION OR TEST	REQUIREMENT PARAGRAPH	INSPECTION METHOD PARAGRAPH
Purity	3.1	4.6.1
Moisture	3.1	4.6.2
Acidity	3.1	4.6.3
pH	3.1	4.6.4
Residue	3.1	4.6.5
Particle Size	3.1	4.6.6
Granulation	3.1	4.6.7
Workmanship	3.1	4.6.8

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of a single production batch of not more than 1000 lbs (454Kg) which has been subjected to the same unit chemical or physical process intended to yield a substantially uniform final product. The lot shall be submitted for inspection in accordance with MIL-STD-105 (or MIL-STD-1235 when applicable).

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4.4.2 Examination. Sampling plans and procedures for the classifications of defects shall be in accordance with MIL-STD-105, except that inspection for critical defects shall be 100 percent. Contractor's sampling plans, if used shall be approved by the government and shall provide, as a minimum, the protection afforded the government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

4.5 Testing.

PRECAUTION WARNING

This specification covers sampling, testing and use of toxic or hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1 Sampling. Using a cement-sampling thief a representative sample of about two pounds in accordance with ASTM Designation Method E300 for solids shall be taken from a container randomly chosen from those comprising the lot.

4.5.2 Acceptance/rejection procedures. The lot represented by the sample shall be tested for compliance with the requirements specified in Table I. The lot shall be accepted if all of the test results comply with the requirements given in Table I. If the test results fail to meet any of the requirements the entire lot shall be rejected.

4.5.3 Classification of defects. The classification of defects shall be as given in Table III.

TABLE III. Classification of defects

TEST	CLASSIFICATION
Purity	Major defect
Moisture	Major defect
Acidity	Major defect
pH	Major defect
Residue	Major defect
Particle size	Major defect
Granulation	Major defect

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4.5.4 Inspection equipment. Commercially available inspection equipment shall be used for all examinations and tests specified in 4.4 and 4.6. The contractor is responsible to have the necessary equipment available, use it correctly and to assure that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.6 Methods of inspection. The following tests shall be performed using prescribed analytical procedures for replicate determination given in standard analytical textbooks. ASTM Method D2905 may also be used to determine the number of replicates required. Unless otherwise specified herein, all chemicals and reagents used in the test methods shall be ACS Grade in accordance with MIL-STD-1218 or Reagent Grade. Purified and distilled water in accordance with ASTM Designation D1193 shall be used as required in all test procedures.

4.6.1 Purity. Accurately weigh to the nearest 0.1mg approximately 0.35g of oxamide and transfer to a clean 600mL beaker. Add 200mL of 5 percent sodium hydroxide solution, stir, and place on a steam bath for 1 hour. Cool to room temperature and add about 270mL of water. Neutralize with 1:1 sulfuric acid and add 40mL 1:1 sulfuric acid in excess. Titrate with standard 0.2 N potassium permanganate solution, adding the reagent rapidly at first and then drop by drop as the end point is approached. The end point is a pink color which persists for 30 seconds. Calculate the percent oxamide as follows:

$$\text{Percent oxamide} = \frac{4.403AN}{W}$$

Where: A = potassium permanganate solution, mL.
N = normality of potassium permanganate solution.
W = weight of sample on a moisture-free basis, gm.

4.6.2 Moisture (volatiles). Accurately weigh to the nearest 0.1mg approximately 5g of the sample into a tared (with cover) low form 60mm diameter weighing bottle. Place in a 105 $\pm$ 3° oven for three hours, cool in a desiccator and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100A-B}{B}$$

Where: A = the weight of sample before drying.
B = the weight of sample after drying.

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4.6.3 Acidity. Accurately weigh approximately 2g of the sample to the nearest 0.001g and quantitatively transfer to a 400mL beaker. Add about 200mL of water which has been made neutral to phenolphthalein and a teflon coated stirring bar and stir on a magnetic stirrer for 5 ± 1 minutes. Filter the mixture through No. 40 Whatman Filter paper into a 600mL beaker. Wash the residue several times with the neutral water and allow the washings to flow into the 600mL beaker. Add 2 or 3 drops of phenophthalein indicator to the water and titrate to a light pink end point with 0.01N sodium hydroxide which has been standardized to 0.0001 N. Calculate the acidity of the oxamide as follows:

$$\text{Percent acidity (as oxalic acid)} = \frac{4.5AN}{W-WM}$$

Where: A = the volume of sodium hydroxide solution in mL
B = the normality of the sodium hydroxide solution
W = the weight of the oxamide sample
M = the percent moisture of the oxamide expressed as a decimal as determined in 4.6.2

4.6.4 pH. Accurately weigh to the nearest 0.001g approximately 2g of Oxamide sample and transfer to a clean 400mL beaker. Add 100mL of freshly boiled and cooled distilled water and stir for several minutes. Allow the solids to settle to the bottom of the beaker, then filter a portion of the water through No. 40 Whatman filter paper into a 200mL beaker. Determine the pH of the water with a pH meter using glass and calomel electrodes. The pH meter shall be calibrated with a standard buffer solution having a pH in the 6-7 range.

4.6.5 Residue on ignition. Accurately weigh to the nearest 0.1mg a 2g sample of the oxamide into a tared platinum crucible. Place the crucible on a supported wire triangle and heat with a Fisher or Meeker or equal gently at first to a red heat. Heat for an additional 15 minutes after all of the organic matter appears to have been completely burned off. Cool the crucible in a desiccator and weigh. Calculate the percent residue on ignition as follows:

$$\text{Residue} = \frac{100A-B}{W-WM}$$

Where: A = the weight of the crucible and contents after ignition
B = the tare weight of the crucible
W = The weight of sample in g
M = the percent moisture as determined in 4.6.2 expressed as a decimal

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4.6.6 Particle size (type I). The particle size of Type I oxamide shall be determined in accordance with Method 205.1 of MIL-STD-650. The density of oxamide is 1.667g/cc.

4.6.7 Granulation (type II). Nest a No. 30 and a No. 325 US Standard sieves conforming with the requirements of RR-S-366 with the No. 352 sieve on the bottom. Nest a bottom pan below the No. 325 sieve and transfer 100g accurately weighed to the nearest 0.01g in the No. 30 sieve. Place a cover on the No. 30 sieve and shake on a mechanical sieve shaker set to produce 300 ± 15 gyrations and 150 ± 10 striker taps per minute for 5 minutes ± 15 seconds. Transfer the oxamide on the sieves and in the bottom pan to appropriate tared weighing containers and weigh each fraction accurately to the nearest 0.01g and calculate the percent oxamide passing through the No. 30 and the No. 325 sieve as follows:

$$\text{Percent oxamide passing thru No. 30 sieve} = \frac{100-100A}{W}$$

$$\text{Percent oxamide passing thru No. 325 sieve} = \frac{100B}{W}$$

Where: A = the weight of oxamide left on the No. 30 sieve in g
B = the weight of oxamide in the bottom pan in g
W = the weight of the oxamide sample in g

4.6.8 Workmanship. Visually examine the contents of the container sampled and the sample taken for the presence of foreign materials.

5. PACKAGING

5.1 Packing and packaging.

5.1.1 Level A. Oxamide shall be packaged in a heat-sealed polyethylene bag in accordance with PPP-C-301 inside a lined steel drum conforming to MIL-D-6054 or fiber drum conforming to PPP-D-723, type III, grade A. Each drum shall not exceed 200 pounds net. Steel drums shall be with a full open head provided with a twist lock closure. The lid shall have a tubular rubber gasket (see 6.1).

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5.1.2 Level B. Oxamide shall be packaged in a heat-sealed polyethylene bag in accordance with PPP-C-301 inside a lined fiber drum conforming to PPP-D-723, type II, grade A. Each drum shall not exceed 100 pounds net. (See 6.1.)

5.1.3 Level C. Oxamide shall be packed in water tight shipping containers in accordance with PPP-C-301 for shipment from the supply source to the first receiving activity for immediate use. Containers shall comply Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation. (See 6.1.)

5.12 Marking. In addition to any special marking specified, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data.

6.1.1 Acquisition requirements. The procurement document should specify the following:

a. Title, number and date of this specification.

b. Quantity required and delivery schedules.

c. Type oxamide required (see 1.1 and 3.1).

d. First article requirements - sample provisions (see 3.2 and 4.3).

e. Level of packing protection required.

6.2.1 Data requirements. When this specification is used in a procurement contract which incorporates DD Form 1423, acceptance and description sheets prepared in accordance with MIL-STD-1171, DRSAR-P-702-102 or as directed in the contract shall be supplied by the contractor.

6.4 Submission of inspection equipment designs for approval. Submit equipment desi ns, as required to: Commander, ARDC, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801-5001.

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6.5 Equivalent test methods. Prior approval of the contracting officer is required for use of equivalent test methods. A description of the proposed method should be submitted thru the contracting officer to: Commander, ARDC, ATTN: AMSMC-QAR-R (D), Dover, NJ 07801. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.6 Drawings. Drawings listed in Section 2 under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as, US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities are now under the cognizance of ARDC.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-60863A	OXAMIDE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*. _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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